

VARUN SREENIVASAN

[Linkedin](#)

| [Website](#)

| (608)-422-2785

| vsreenivasanofficial@gmail.com

EXPERIENCE

DataChat, Inc.

Senior Software Engineer

Madison, WI

Feb 2022 - Present

- Consistently achieved high ratings across *all* regular performance appraisals.
- Added Google Vector Search & Milvus*: Led team of devs to overhaul VectorDB platform with integration of critical providers.
- Scalability Upgrades*: Proposed & implemented [Artillery](#) driven load tests; uncovered bottlenecks impeding platform functionality.
- Increased Automation & Delivery Cadence*: Led CI/CD push to streamline release mgmt process involving [Kubernetes](#) deployments.
- Accelerated LLM adoption*: Setup evaluation framework for benchmarking LLMs across various use cases; led research assessing feasibility of hosting DataChat's custom LLMs across multiple cloud providers and platforms.
- Expanded GenAI & ML Product Capabilities*: Spearheaded design & implementation of backend solutions; achieved scalability, compute efficiency and reliability with system design best practices and cloud native architecture.

National Science Foundation – IRIS HEP

[Fellow](#) – Graph Methods for Particle Tracking (High Luminosity Large Hadron Collider)

Berkeley, CA

May 2021 - August 2021

- Optimized Particle Differentiation*: Developed embedding pipeline to find distance metric between 3D hit measurements pairs.
- Accelerated Nearest Neighbors Search*: Replaced Facebook's Faiss with Fixed Radius Nearest Neighbors (FRNN) on CUDA.

Citrine Informatics

[NextGen-Fellow](#) – Computational Materials Science

Redwood City, CA

May 2018 - August 2018

- Multi-University Research Project*: Competitively selected, successfully completed bootcamp & workshop at Stanford University, presented findings at NextGen Research Symposium in Golden, CO, and co-authored a paper.
- Optimized Metal Defect Detection*: Employed RetinaNet model; developed evaluation pipeline to determine recall and precision metrics; obtained model with 85% precision and 68% recall on test set with parameter tuning.

EDUCATION

M.S. Computer Science, University of Wisconsin-Madison (Dec 2021)

G.P.A: 4.00/4.00

B.S. Computer Science (Minor: Mathematics), University of Wisconsin-Madison (May 2020)

G.P.A: 3.90/4.00 "Distinction in the Major" and Dean's List

LANGUAGES & TECHNOLOGY SKILLSETS

Python, Git, AWS, GCP, Azure, Terraform, Docker, Kubernetes, Helm, CircleCI, VS Code, Java, Jupyter Notebook, Unix, MacOS

LICENSES & CERTIFICATIONS

Amazon Web Services: Solutions Architect Associate, Cloud Practitioner, Machine Learning Specialty

OTHER SOFTWARE R&D PROJECTS

Autonomous Vehicles

Live Traffic-Sign detection: Trained PyTorch MobileNet detection model; developed proof of concept for open source testbed; presented proposal at ACM MobiArch'22.

Instance Segmentation App

Segmentation With User Images: Created app that accurately identifies and separates objects of interest from background.

Business Success & Viability Forecast - Covid

Developed Multiple ML models with Yelp Dataset: Architected feature engineering, created training and validation pipelines, and determined vital factors to predict business survival during Covid.

RELEVANT COURSEWORK

Machine Learning, Data Science, Artificial Intelligence, Algorithms, Computer Networks, Mobile & Wireless Networks, Computer Vision, Cryptography, Combinatorics, Linear Algebra & Differential Equations, Operating Systems.

PUBLICATIONS & PRESENTATIONS

- [μCity: a miniaturized autonomous vehicle testbed](#): Procs. 17th ACM Workshop **MobiArch'22**
- [Multi defect detection & analysis of electron microscopy images with deep learning](#): Science Direct (Computational Material Sci.)
- [Graph methods for particle tracking](#): IRIS HEP fellowship final presentation